

JONATHAN D. SHAW
✉: jonathan.shaw@colorado.edu
🌐: <https://jonathan-d-shaw.github.io>

Education

Ph.D. in Applied Mathematics
University of Colorado Boulder

Expected May 2028
Boulder, CO

M.S. in Applied Mathematics
University of Colorado Boulder

May 2026
Boulder, CO

B.S. in Applied Mathematics
University of Colorado Boulder
Magna cum laude
Concentration in Aerospace Engineering

May 2023
Boulder, CO

Research Interests

- Numerical partial differential equations
- Integral equations
- “Fast” algorithm design
- Multiscale modeling/simulation
- Scattering and inverse problems
- Applications of functional, asymptotic, and numerical analysis

Publications

1. Trujillo, M.; Sajadi, A.; **Shaw, J. D.**; Hodge, B.-M., *Computationally Efficient Analytical Models of Frequency and Voltage in Low-Inertia Systems*, IEEE Transactions on Power Delivery 2026 (🌐: [arXiv](#))
2. Trujillo, M.; Sajadi, A.; **Shaw, J. D.**; Hodge, B.-M., *Analytical Models of Frequency and Voltage in Large-Scale All-Inverter Power Systems*, IEEE Transactions on Power Systems 2025 (🌐: [arXiv](#))
3. **Shaw, J. D.**; Restrepo, J. G.; Rodríguez, N., *A dynamical model of gentrification: Exploring a simple rent control strategy*, Phys. Rev. E, 2024 (🌐: [arXiv](#))

Awards

- Student Travel Award, Dynamics Days US 2025

Talks & Presentations

Talks

- *Improving the density interpolation method*, SIAM Graduate Student Chapter Seminar, University of Colorado Boulder, January 2026 (contributed talk)
- *A dynamical system model of gentrification*, Complex/Dynamical Systems Seminar, University of Colorado Boulder, March 2025 (invited talk)
- *A dynamical system model of gentrification*, Dynamics Days US 2025, Denver, CO, January 2025 (contributed talk)

- *On the dynamics of gentrification*, Mathematical Biology Seminar, University of Colorado Boulder, October 2023 (contributed talk)

Poster Presentations

- *A dynamical system model of gentrification*, Dynamics Days US 2023, Denver, CO, January 2023

Research Experience

Department of Applied Mathematics	University of Colorado Boulder
<i>Research Assistant</i>	<i>May 2022–Present</i>

QUASI-DIRECT SOLVERS FOR BOUNDARY INTEGRAL EQUATIONS	<i>May 2026–Present</i>
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Project advisor: [Adrianna Gillman](#)

IMPROVING THE <i>Density Interpolation Method</i>	<i>August 2025–May 2026</i>
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Project advisor: [Eduardo Corona](#)

FAST SOLVERS FOR POWER SYSTEMS APPLICATIONS	<i>May 2023–August 2025</i>
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Project advisor: [Eduardo Corona](#)

Project collaborators: [Fiona Majeau](#), [Bri-Mathias Hodge](#), [Amir Sajadi](#)

GENTRIFICATION DYNAMICS	<i>May 2022–September 2024</i>
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Project advisors: [Juan Restrepo](#) and [Nancy Rodríguez](#)

Los Alamos National Laboratory	Los Alamos, NM
<i>Intern - Physics XTD-PRI</i>	<i>May 2018–August 2018</i>

ALGEBRAIC TOPOLOGY FOR COMPUTATIONAL FLUID DYNAMICS	<i>May 2018–August 2018</i>
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Project advisor: [Roy S. Baty](#)

Software

Contributions

- [chunkIE](#): A MATLAB integral equation toolbox

Leadership & Service

Society for Industrial and Applied Mathematics (SIAM)	University of Colorado Boulder
<i>President, Graduate Student Chapter</i>	<i>August 2025–Present</i>

<i>Industry Liaison, Graduate Student Chapter</i>	<i>August 2024–May 2025</i>
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Buckskin Youth Leadership Training	Richmond, VA
<i>Senior Staff Member</i>	<i>January 2016–August 2023</i>